



PROPOSED PACKAGE INSERT  
PRODUCT NAME: Seacross Docetaxel 20mg/ml Concentrate for Solution for Infusion  
ACTIVE INGREDIENT: DOCETAXEL 20mg/ml

PRODUCT DESCRIPTION

Before and After Dilution: Docetaxel 20 mg/ml concentrate for solution for infusion is colourless to either yellowish or greenish-yellow solution.

It is available in three presentations:

Docetaxel 20 mg/1 ml, 80 mg/4 ml and 160mg/8ml concentrate for solution for infusion.  
Compatible with 5% glucose solution or 0.9% sodium chloride solution.

PHARMACODYNAMICS

**Mechanism of Action:** Docetaxel is an antineoplastic agent which acts by promoting the assembly of tubulin into stable microtubules and inhibits their disassembly which leads to a marked decrease of free tubulin. The binding of docetaxel to microtubules does not alter the number of protofilaments.

Docetaxel has been shown *in vitro* to disrupt the microtubular network in cells which is essential for vital mitotic and interphase cellular functions.

**Pharmacodynamic Effects:** Docetaxel was found to be cytotoxic *in vitro* against various murine and human tumour cell lines and against freshly excised human tumour cells in clonogenic assays. Docetaxel achieves high intracellular concentrations with a long cell residence time. In addition, docetaxel was found to be active on some but not all cell lines over expressing the p-glycoprotein which is encoded by the multidrug resistance gene.

*In vivo*, docetaxel is schedule independent and has a broad spectrum of experimental antitumour activity against advanced murine and human grafted tumours.

PHARMACOKINETICS

**Absorption:** The kinetic profile of docetaxel is dose-independent and consistent with a 3-compartment pharmacokinetic model with half lives for the  $\alpha$ -,  $\beta$ - and  $\gamma$ -phases of 4 min, 36 min and 11.1 hrs, respectively. The late phase is due, in part, to a relatively slow efflux of docetaxel from the peripheral compartment.

**Distribution:** Following the administration of a 100-mg/m<sup>2</sup> dose given as a 1-hr infusion a mean peak plasma level of 3.7 mcg/mL was obtained with a corresponding AUC of 4.6 hr-mcg/mL. Mean values for total body clearance and steady-state volume of distribution were 21 L/hr/m<sup>2</sup> and 113 L, respectively. Inter-individual variation in total body clearance was approximately 50%. Docetaxel is >95% bound to plasma proteins.

**Elimination:** Docetaxel was eliminated in both the urine and faeces following cytochrome P-450-mediated oxidative metabolism of the tert-butyl ester group, within 7 days, the urinary and faecal excretion accounted for about 6% and 75% of the administered radioactivity, respectively. About 80% of the radioactivity recovered in faeces is excreted during the first 48 hrs as 1 major inactive metabolite and 3 minor inactive metabolites and very low amounts of unchanged medicinal product.

**Special Populations: Age and Gender:** The pharmacokinetics of docetaxel were not altered by the age or sex of the patient.

**Hepatic Impairment:** In a small number of patients suggestive of mild to moderate liver function impairment (ALT, AST  $\geq$  1.5 times the ULN associated with alkaline phosphatase  $\geq$  2.5 times the ULN), total clearance was lowered by 27% on average (see Dosage & Administration).

**Fluid Retention:** Docetaxel clearance was not modified in patients with mild to moderate fluid retention and there are no data available in patients with severe fluid retention.

**Combination Therapy: Doxorubicin:** When used in combination, docetaxel does not influence the clearance of doxorubicin and the plasma levels of doxorubicinol (a doxorubicin metabolite). The pharmacokinetics of docetaxel, doxorubicin and cyclophosphamide were not influenced by their co-administration.

**Capecitabine:** No effect by capecitabine on the pharmacokinetics of docetaxel [peak plasma concentration (C<sub>max</sub>) and area under the concentration-time curve (AUC)] and no effect by docetaxel on the pharmacokinetics of a relevant capecitabine metabolite 5'-DFUR.

**Cisplatin:** Clearance of docetaxel in combination therapy with cisplatin was similar to that observed following monotherapy. The pharmacokinetic profile of cisplatin administered shortly after docetaxel infusion is similar to that observed with cisplatin alone.

**Cisplatin and 5-Fluorouracil:** The combined administration of docetaxel, cisplatin and 5-fluorouracil had no influence on the pharmacokinetics of each individual medicinal product.

**Prednisone and Dexamethasone:** The effect of prednisone on the pharmacokinetics of docetaxel administered with standard dexamethasone premedication shows no effect of prednisone on the pharmacokinetics of docetaxel was observed.

**Toxicology: Preclinical Safety Data:** The carcinogenic potential of docetaxel has not been studied.

Docetaxel has been shown to be mutagenic in the *in vitro* micronucleus and chromosome aberration test in CHO-K1 cells and in the *in vivo* micronucleus test in the mouse. However, it did not induce mutagenicity in the Ames test or the CHO/HGPRT gene mutation assay. These results are consistent with the pharmacological activity of docetaxel.

Undesirable effects on the test is observed in rodent toxicity studies suggest that docetaxel may impair male fertility.

INDICATION

**Breast Cancer:** Docetaxel in combination with doxorubicin and cyclophosphamide is indicated for the adjuvant treatment of patients with operable node-positive breast cancer.

- Docetaxel in combination with doxorubicin is indicated for the treatment of patients with locally advanced or metastatic breast cancer who have not previously received cytotoxic therapy for this condition.
- Docetaxel in combination with capecitabine is indicated for the treatment of patients with locally advanced or metastatic breast cancer after failure of cytotoxic chemotherapy. Previous therapy should have included an anthracycline.
- Docetaxel monotherapy is indicated for the treatment of patients with locally advanced or metastatic breast cancer in whom previous therapy has failed. Prior therapy should have included an anthracycline unless clinically contraindicated.

**Non-Small Cell Lung Cancer:** Docetaxel is indicated for the treatment of patients with locally advanced or metastatic non-small cell lung cancer even after failure of platinum-based chemotherapy.

- Docetaxel in combination with cisplatin is indicated for the treatment of patients with unresectable, locally advanced or metastatic non-small cell lung cancer, in patients who have not previously received chemotherapy for this condition. Docetaxel in combination with carboplatin represents a treatment option to cisplatin-based therapy.

**Ovarian Cancer:** Docetaxel is indicated for the treatment of metastatic carcinoma of the ovary after failure of first-line or subsequent chemotherapy.

**Squamous Cell Carcinoma of the Head and Neck:** Docetaxel is indicated as monotherapy in the treatment of patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck after failure of a previous chemotherapy regimen.

**Prostate Cancer:** Docetaxel in combination with prednisone or prednisolone is indicated for the treatment of patients with hormone refractory metastatic prostatic cancer.

**Gastric Adenocarcinoma:** Docetaxel in combination with cisplatin and 5-fluorouracil is indicated for the treatment of patients with metastatic gastric adenocarcinoma, including adenocarcinoma of the gastroesophageal junction, who have not received prior chemotherapy for metastatic disease.

**Head and neck cancer:** Docetaxel in combination with cisplatin and 5-fluorouracil is indicated for the induction treatment of patients with locally advanced squamous cell carcinoma of the head and neck.

The use of docetaxel should be confined to units specialised in the administration of cytotoxic chemotherapy and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy.

POSOLGY

Recommended dose

For breast, non-small cell lung, gastric, and head and neck cancers, premedication consisting of an oral corticosteroid, such as dexamethasone 16 mg per day (e.g. 8 mg BID) for 3 days starting 1 day prior to docetaxel administration, unless contraindicated, can be used.

For prostate cancer, given the concurrent use of prednisone or prednisolone the recommended premedication regimen is oral dexamethasone 8 mg, 12 hours, 3 hours and 1 hour before the docetaxel infusion.

Prophylactic G CSF may be used to mitigate the risk of haematological toxicities.

Docetaxel is administered as a one-hour infusion every three weeks.

Breast cancer

In the adjuvant treatment of operable node-positive breast cancer, the recommended dose of docetaxel is 75 mg/m<sup>2</sup> administered 1-hour after doxorubicin 50 mg/m<sup>2</sup> and cyclophosphamide 500 mg/m<sup>2</sup> every 3 weeks for 6 cycles (see also Dose adjustments during treatment). For the treatment of patients with locally advanced or metastatic breast cancer, the recommended dose of docetaxel is 100 mg/m<sup>2</sup> in monotherapy. In first-line treatment, docetaxel 75 mg/m<sup>2</sup> is given in combination therapy with doxorubicin (50 mg/m<sup>2</sup>).

In combination with capecitabine, the recommended dose of docetaxel is 75 mg/m<sup>2</sup> every three weeks, combined with capecitabine at 1250 mg/m<sup>2</sup> twice daily (within 30 minutes after a meal) for 2 weeks followed by 1-week rest period. For capecitabine dose calculation according to body surface area, see capecitabine summary of product characteristics.

Non-small cell lung cancer

In chemotherapy naive patients treated for non-small cell lung cancer, the recommended dose regimen is docetaxel 75 mg/m<sup>2</sup> immediately followed by cisplatin 75 mg/m<sup>2</sup> over 30-60 minutes. For treatment after failure of prior platinum-based chemotherapy, the recommended dose is 75 mg/m<sup>2</sup> as a single agent.

Prostate cancer

The recommended dose of docetaxel is 75 mg/m<sup>2</sup>. Prednisone or prednisolone 5 mg orally twice daily is administered continuously.

Gastric adenocarcinoma

The recommended dose of docetaxel is 75 mg/m<sup>2</sup> as a 1 hour infusion, followed by cisplatin 75mg/m<sup>2</sup>, as a 1 to 3 hour infusion (both on day 1 only), followed by 5-fluorouracil 750 mg/m<sup>2</sup> per day given as a 24-hour continuous infusion for 5 days, starting at the end of the cisplatin infusion. Treatment is repeated every three weeks. Patients must receive premedication with antiemetics and appropriate hydration for cisplatin administration. Prophylactic G-CSF should be used to mitigate the risk of hematological toxicities (see also Dose adjustments during treatment).

Ovarian Cancer

The recommended dosage is 100 mg/m<sup>2</sup> administered as one hour infusion every three weeks. When use in combination, docetaxel is administered at the recommended dosage of 75 mg/m<sup>2</sup>.

Squamous Cell Carcinoma of the Head and Neck

The recommended dosage is 100 mg/m<sup>2</sup> administered as one hour infusion every three weeks. When use in combination, docetaxel is administered at the recommended dosage of 75 mg/m<sup>2</sup>.

Patients must receive premedication with antiemetics and appropriate hydration (prior to and after cisplatin administration). Prophylactic G-CSF may be used to mitigate the risk of hematological toxicities. All patients on the docetaxel-containing arm, received prophylactic antibiotics.

- Induction chemotherapy followed by radiotherapy

For the induction treatment of inoperable locally advanced squamous cell carcinoma of the head and neck (SCCHN), the recommended dose of docetaxel is 75 mg/m<sup>2</sup> as a 1 hour infusion followed by cisplatin 75 mg/m<sup>2</sup> over 1 hour, on day one, followed by 5-fluorouracil as a continuous infusion at 150 mg/m<sup>2</sup> per day for five days. This regimen is administered every 3 weeks for 4 cycles. Following chemotherapy, patients should receive radiotherapy.

- Induction chemotherapy followed by chemoradiotherapy

For the induction treatment of patients with locally advanced (technically unresectable, low probability of surgical cure, and aiming at organ preservation) squamous cell carcinoma of the head and neck (SCCHN), the recommended dose of docetaxel is 75 mg/m<sup>2</sup> as a 1 hour intravenous infusion on day 1, followed by cisplatin 100 mg/m<sup>2</sup> administered as a 30-minute to 1 hour infusion, followed by 5-fluorouracil 1000 mg/m<sup>2</sup>/day as a continuous infusion from day 1 to day 4. This regimen is administered every 3 weeks for 3 cycles. Following chemotherapy, patients should receive chemoradiotherapy.

For cisplatin and 5-fluorouracil dose modifications, see the corresponding summary of product characteristics.

Dose adjustments during treatment

General

Docetaxel should be administered when the neutrophil count is 1,500 cells/mm<sup>3</sup>. In patients who experienced either febrile neutropenia, neutrophil < 500 cells/mm<sup>3</sup> for more than one week, severe or cumulative cutaneous reactions or severe peripheral neuropathy during docetaxel therapy, the dose of docetaxel should be reduced from 100 mg/m<sup>2</sup> to 75 mg/m<sup>2</sup> and/or from 75 to 60 mg/m<sup>2</sup>, if the patient continues to experience these reactions at 60 mg/m<sup>2</sup>, the treatment should be discontinued.

Adjuvant therapy for breast cancer

In the pivotal trial in patients who received adjuvant therapy for breast cancer and who experienced complicated neutropenia (including prolonged neutropenia, febrile neutropenia, or infection), it was recommended to use G-CSF to provide prophylactic coverage (eg, day 4 to 11) in all subsequent cycles. Patients who continued to experience this reaction should remain on G-CSF and have their docetaxel dose reduced to 60 mg/m<sup>2</sup>.

However, in clinical practice neutropenia could occur earlier. Thus the use of G-CSF should be considered function of the neutropenic risk of the patient and current recommendations. Patients who experience Grade 3 or 4 stomatitis should have their dose decreased to 60mg/m<sup>2</sup>.

In combination with cisplatin

For patients who are dosed initially at docetaxel 75 mg/m<sup>2</sup> in combination with cisplatin and whose nadir of platelet count during the previous course of therapy is < 25,000 cells/mm<sup>3</sup>, or in patients who experience febrile neutropenia, or in patients with serious non-hematologic toxicities, the docetaxel dose in subsequent cycles should be reduced to 65 mg/m<sup>2</sup>. For cisplatin dose adjustments, see the corresponding summary of product characteristics.

In combination with capecitabine

- For capecitabine dose modifications, see capecitabine summary of product characteristics.
- For patients developing the first appearance of a Grade 2 toxicity, which persists at the time of the next docetaxel/capecitabine treatment, delay treatment until resolved to Grade 0-1, and resume at 100% of the original dose.
- For patients developing the second appearance of a Grade 2 toxicity, or the first appearance of a Grade 3 toxicity, at any time during the treatment cycle, delay treatment until resolved to Grade 0-1, then resume treatment with docetaxel 55 mg/m<sup>2</sup>.
- For any subsequent appearances of toxicities, or any Grade 4 toxicities, discontinue the docetaxel dose.

In combination with cisplatin and 5-fluorouracil

If an episode of febrile neutropenia, prolonged neutropenia or neutropenic infection occurs despite G-CSF use, the docetaxel dose should be reduced from 75 to 60 mg/m<sup>2</sup>. If subsequent episodes of complicated neutropenia occur the docetaxel dose should be reduced from 60 to 45 mg/m<sup>2</sup>. In case of Grade 4 thrombocytopenia the docetaxel dose should be reduced from 75 to 60 mg/m<sup>2</sup>. Patients should not be retreated with subsequent cycles of docetaxel until neutrophils recover to a level >1,500 cells/mm<sup>3</sup> and platelets recover to a level >100,000 cells/mm<sup>3</sup>. Discontinue treatment if these toxicities persist (see section Special warnings and precautions for use).

Recommended dose modifications for toxicities in patients treated with docetaxel in combination with cisplatin and 5-fluorouracil (5-FU):

| Toxicity                     | Dose adjustment                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diarrhoea grade 3            | First episode: reduce 5-FU dose by 20%.<br>Second episode: then reduce docetaxel dose by 20%.                                                        |
| Diarrhoea grade 4            | First episode: reduce docetaxel and 5-FU doses by 20%.<br>Second episode: discontinue treatment.                                                     |
| Stomatitis/mucositis grade 3 | First episode: reduce 5-FU dose by 20%.<br>Second episode: stop 5-FU only, at all subsequent cycles.<br>Third episode: reduce docetaxel dose by 20%. |
| Stomatitis/mucositis grade 4 | First episode: stop 5-FU only, at all subsequent cycles.<br>Second episode: reduce docetaxel dose by 20%.                                            |

For cisplatin and 5-fluorouracil dose adjustments, see the corresponding summary of product information.

Special populations:

Patients with hepatic impairment

Based on pharmacokinetic data with docetaxel at 100 mg/m<sup>2</sup> as single agent, patients who have both elevations of transaminase (ALT and/or AST) greater than 1.5 times the upper limit of the normal range (ULN) and alkaline phosphatase greater than 2.5 times the ULN, the recommended dose of docetaxel is 75 mg/m<sup>2</sup> (see sections Special warnings and precautions for use and Pharmacokinetic properties). For those patients with serum bilirubin > ULN and/or ALT and AST > 3.5 times the ULN associated with alkaline phosphatase > 6 times the ULN, no dose-reduction can be recommended and docetaxel should not be used unless strictly indicated. In combination with cisplatin and 5-fluorouracil for the treatment of patients with gastric adenocarcinoma, the pivotal clinical study excluded patients with ALT and/or AST > 1.5 x ULN associated with alkaline phosphatase > 2.5 x ULN, and bilirubin > 1 x ULN; for these patients, no dose-reductions can be recommended and docetaxel should not be used unless strictly indicated. No data are available in patients with hepatic impairment treated by docetaxel in combination in the other indications.

Paediatric population

The safety and efficacy of Docetaxel in nasopharyngeal carcinoma in children aged 1 month to less than 18 years have not yet been established. There is no relevant use of Docetaxel in the paediatric population in the indications breast cancer, non-small cell lung cancer, prostate cancer, gastriccarcinoma and head and neck cancer, not including type II and III less differentiated nasopharyngeal carcinoma.

Older People

Based on a population pharmacokinetic analysis, there are no special instructions for use in the elderly. In combination with capecitabine, for patients 60 years of age or more, a starting dose reduction of capecitabine to 75% is recommended (see capecitabine summary of product characteristics).

Special precautions for disposal and other handling

Docetaxel is an antineoplastic agent and, as with other potentially toxic compounds, caution should be exercised when handling it and preparing Docetaxel solutions. The use of gloves is recommended.

If Docetaxel concentrate or infusion solution should come into contact with skin, wash immediately and thoroughly with soap and water. If Docetaxel concentrate or infusion solution should come into contact with mucous membranes, wash immediately and thoroughly with water.

Preparation of the infusion solution

**Docetaxel 20 mg/ml concentrate for solution for infusion requires NO prior dilution with a solvent and is ready to add to the infusion solution.**

Each vial is of single use and should be used immediately. More than one vial of Docetaxel concentrate for solution for infusion may be necessary to obtain the required dose for the patient. Aseptically withdraw the required amount of Docetaxel concentrate for solution for infusion using a calibrated syringe. The required volume of Docetaxel concentrate for solution for infusion must be injected into a 250 ml infusion bag or bottle containing either glucose solution 5% or sodium chloride 0.9% solution for infusion. If a dose greater than 190 mg of docetaxel is required, use a larger volume of the infusion vehicle so that a concentration of 0.74 mg/ml docetaxel is not exceeded. Mix the infusion bag or bottle manually using a rocking motion. As with all parenteral products, Docetaxel infusion solution should be visually inspected prior to use, solutions containing a precipitate should be discarded. Any unused product or waste material should be disposed of in accordance with local requirements.

ROUTE OF ADMINISTRATION:INTRAVENOUS

CONTRAINDICATION

Hypersensitivity to docetaxel or to any of the excipients of Docetaxel.

Patients with baseline neutrophil count of <1500 cells/mm<sup>3</sup> and those with severe liver impairment (see Dosage & Administration, Warnings and Precautions).

Contraindications for other medicinal products also apply, when combined with docetaxel.

WARNING AND PRECAUTIONS

For breast and non-small cell lung cancers, premedication consisting of an oral corticosteroid, such as dexamethasone 16 mg per day (e.g. 8 mg BID) for 3 days starting 1 day prior to docetaxel administration, unless contraindicated, can reduce the incidence and severity of fluid retention aswell as the severity of hypersensitivity reactions. For prostate cancer, the premedication is oral dexamethasone 8 mg, 12 hours, 3 hours and 1 hour before the docetaxel infusion.



## Haematology

Neutropenia is the most frequent adverse reaction of docetaxel. Neutrophil nadirs occurred at a median of 7 days but this interval may be shorter in heavily pre-treated patients. Frequent monitoring of complete blood counts should be conducted on all patients receiving docetaxel. Patients should be retreated with docetaxel when neutrophils recover to a level  $\geq 1,500$  cells/mm<sup>3</sup>.

In the case of severe neutropenia (< 500 cells/mm<sup>3</sup> for seven days or more) during a course of docetaxel therapy, a reduction in dose for subsequent courses of therapy or the use of appropriate symptomatic measures are recommended.

In patients treated with docetaxel in combination with cisplatin and 5-fluorouracil (TCF), febrile neutropenia and neutropenic infection occurred at lower rates when patients received prophylactic G-CSF. Patients treated with TCF should receive prophylactic G-CSF to mitigate the risk of complicated neutropenia (febrile neutropenia, prolonged neutropenia or neutropenic infection). Patients receiving TCF should be closely monitored.

In patients treated with docetaxel in combination with doxorubicin and cyclophosphamide (TAC), febrile neutropenia and/or neutropenic infection occurred at lower rates when patients received primary G-CSF prophylaxis. Primary G-CSF prophylaxis should be considered in patients who receive adjuvant therapy with TAC for breast cancer to mitigate the risk of complicated neutropenia (febrile neutropenia, prolonged neutropenia or neutropenic infection). Patients receiving TAC should be closely monitored.

## Hypersensitivity reactions

Patients should be observed closely for hypersensitivity reactions especially during the first and second infusions. Hypersensitivity reactions may occur within a few minutes following the initiation of the infusion of docetaxel, thus facilities for the treatment of hypotension and bronchospasm should be available. If hypersensitivity reactions occur, minor symptoms such as flushing or localised cutaneous reactions do not require interruption of therapy. However, severe reactions, such as severe hypotension, bronchospasm or generalised rash/erythema require immediate discontinuation of docetaxel and appropriate therapy. Patients who have developed severe hypersensitivity reactions should not be re-challenged with docetaxel.

## Cutaneous reactions

Localised skin erythema of the extremities (palms of the hands and soles of the feet) with oedema followed by desquamation has been observed. Severe symptoms such as eruptions followed by desquamation which lead to interruption or discontinuation of docetaxel treatment were reported.

## Fluid retention

Patients with severe fluid retention such as pleural effusion , pericardial effusion and ascites should be monitored closely.

## Respiratory disorders

Acute respiratory distress syndrome, interstitial pneumonia/pneumonitis, interstitial lung disease, pulmonary fibrosis and respiratory failure have been reported and may be associated with fatal outcome. Cases of radiation pneumonitis have been reported in patients receiving concomitant radiotherapy.

If new or worsening pulmonary symptoms develop, patients should be closely monitored, promptly investigated, and appropriately treated. Interruption of docetaxel therapy is recommended until diagnosis is available. Early use of supportive care measures may help improve the condition. The benefit of resuming docetaxel treatment must be carefully evaluated.

## Patients with liver impairment

In patients treated with docetaxel at 100 mg/m<sup>2</sup> as single agent who have serum transaminase levels (ALT and/or AST) greater than 1.5 times the ULN concurrent with serum alkaline phosphatase levels greater than 2.5 times the ULN, there is a higher risk of developing severe adverse reactions such as toxic deaths including sepsis and gastrointestinal haemorrhage which can be fatal, febrile neutropenia, infections, thrombocytopenia, stomatitis and asthenia. Therefore, the recommended dose of docetaxel in those patients with elevated liver function test (LFTs) is 75 mg/m<sup>2</sup> and LFTs should be measured at baseline and before each cycle.

For patients with serum bilirubin levels > ULN and/or ALT and AST > 3.5 times the ULN concurrent with serum alkaline phosphatase levels > 6 times the ULN, no dose-reduction can be recommended and docetaxel should not be used unless strictly indicated.

In combination with cisplatin and 5-fluorouracil for the treatment of patients with gastric adenocarcinoma, the pivotal clinical study excluded patients with ALT and/or AST > 1.5 × ULN associated with alkaline phosphatase > 2.5 × ULN, and bilirubin > 1 x ULN; for these patients, no dose-reductions can be recommended and docetaxel should not be used unless strictly indicated. No data are available in patients with hepatic impairment treated by docetaxel in combination in the other indications.

## Patients with renal impairment

There are no data available in patients with severely impaired renal function treated with docetaxel.

## Nervous system

The development of severe peripheral neurotoxicity requires a reduction of dose.

## Cardiac toxicity

Heart failure has been observed in patients receiving docetaxel in combination with trastuzumab, particularly following anthracycline (doxorubicin or epirubicin)-containing chemotherapy. This may be moderate to severe and has been associated with death.

When patients are candidates for treatment with docetaxel in combination with trastuzumab, they should undergo baseline cardiac assessment. Cardiac function should be further monitored during treatment (e.g. every three months) to help identify patients who may develop cardiac dysfunction. For more details see summary of product characteristics of trastuzumab.

## Eye disorders

Cystoid macular oedema (CMO) has been reported in patients treated with docetaxel. Patients with impaired vision should undergo a prompt and complete ophthalmologic examination. In case CMO is diagnosed, docetaxel treatment should be discontinued and appropriate treatment initiated.

## Others

Contraceptive measures must be taken by both men and women during treatment and for men at least 6 months after cessation of therapy.

The concomitant use of docetaxel with strong CYP3A4 inhibitors (e.g., ketoconazole, itraconazole, clarithromycin, indinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin and voriconazole) should be avoided.

Additional cautions for use in adjuvant treatment of breast cancer

## Complicated neutropenia

For patients who experience complicated neutropenia (prolonged neutropenia, febrile neutropenia or infection), G-CSF and dose reduction should be considered.

## Gastrointestinal reactions

Symptoms such as early abdominal pain and tenderness, fever, diarrhoea, with or without neutropenia, may be early manifestations of serious gastrointestinal toxicity and should be evaluated and treated promptly.

## Congestive heart failure (CHF)

Patients should be monitored for symptoms of congestive heart failure during therapy and during the follow up period. In patients treated with the TAC regimen for node positive breast cancer, the risk of CHF has been shown to be higher during the first year after treatment.

## Patients with 4+ nodes

As the benefit observed in patient with 4+ nodes was not statistically significant on disease-free survival (DFS) and overall survival (OS), the positive benefit/risk ratio for TAC in patients with 4+ nodes was not fully demonstrated at the final analysis.

## Older people

There are limited data available in patients > 70 years of age on docetaxel use in combination with doxorubicin and cyclophosphamide.

In patients treated with docetaxel every three weeks, the incidence of related nail changes occurred at a rate  $\geq 10\%$  higher in patients who were 65 years of age or greater compared to younger patients. The incidence of related fever, diarrhoea, anorexia, and peripheral oedema occurred at rates  $\geq 10\%$  higher in patients who were 75 years of age or greater versus less than 65 years.

Among the patients treated with docetaxel in combination with cisplatin and 5-fluorouracil in the gastric cancer, the incidence of serious adverse events was higher in older people compared to younger patients. The incidence of the following adverse events (all grades): lethargy, stomatitis, neutropenic infection occurred at rates  $\geq 10\%$  higher in patients who were 65 years of age or older compared to younger patients.

Older people treated with TCF should be closely monitored.

## Excipients

This medicinal product contains approximately 55 vol % ethanol (alcohol), i.e. up to 395 g (0.5 ml) ethanol anhydrous per 1 ml vial, equivalent to 10 ml of beer or 4 ml wine per 1 ml vial, 1.58 g (2 ml) ethanol anhydrous per 4 ml vial, equivalent to 40 ml of beer or 17 ml wine per 4 ml vial, or 3.16g (4 ml) ethanol anhydrous per vial of 8 ml fill volume, equivalent to 80 ml beer or 33ml wine per 8 ml vial.

Harmful for those suffering from alcoholism.

To be taken into account in pregnant or breast-feeding women, children and high-risk groups such as patients with liver disease, or epilepsy.

Consideration should be given to possible effects on the central nervous system. The amount of alcohol in this medicinal product may alter the effects of other medicinal products. The amount of alcohol in this medicinal product may impair the patient's ability to drive or use machines.

Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. The amount of ethanol in Seacross Docetaxel may impair the ability to drive or use machines.

## INTERACTIONS WITH OTHER MEDICAMENTS

The metabolism of docetaxel may be modified by the concomitant administration of compounds which induce, inhibit or are metabolised by (and thus may inhibit the enzyme competitively) cytochrome P450-3A such as ciclosporine, ketoconazole and erythromycin. As a result, caution should be exercised when treating patients with these medicinal products as concomitant therapy since there is a potential for a significant interaction.

In case of combination with CYP3A4 inhibitors, the occurrence of docetaxel adverse reactions may increase, as a result of reduced metabolism. If the concomitant use of a strong CYP3A4 inhibitor (e.g., ketoconazole, itraconazole, clarithromycin, indinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin and voriconazole) cannot be avoided, a close clinical surveillance is warranted and a dose-adjustment of docetaxel may be suitable during the treatment with the strong CYP3A4 inhibitor. The co-administration of docetaxel with the strong CYP3A4 inhibitor ketoconazole leads to a significant decrease in docetaxel clearance by 49%.

Docetaxel is metabolised by CYP3A4 and prednisone is known to induce CYP3A4. Docetaxel pharmacokinetics in the presence of prednisone has showed no statistically significant effect of prednisone on the pharmacokinetics of docetaxel was observed. Docetaxel is highly protein bound (> 95%). Although the possible in vivo interaction of docetaxel

with concomitantly administered medicinal product has not been investigated formally, in vitro interactions with lightly protein-bound agents such as erythromycin, diphenhydramine, propranolol, propafenone, phenytoin, salicylate, sulfamethoxazole and sodium valproate did not affect protein binding of docetaxel. In addition, dexamethasone did not affect protein binding of docetaxel. Docetaxel did not influence the binding of digitoxin.

The pharmacokinetics of docetaxel, doxorubicin and cyclophosphamide were not influenced by their co-administration. Limited data were suggestive of an interaction between docetaxel andcarboplatin. When combined to docetaxel, the clearance of carboplatin was about 50% higher than values previously reported for carboplatin monotherapy.

## PREGNANCY AND LACTATION

Use in pregnancy: There is no information on the use of docetaxel in pregnant women.

Docetaxel has been shown to be both embryotoxic and foetotoxic in rabbits and rats, and to reduce fertility in rats. As with other cytotoxic medicinal products, docetaxel may cause foetal harm when administered to pregnant women. Therefore, docetaxel must not be used during pregnancy unless clearly indicated.

Women of childbearing age receiving docetaxel should be advised to avoid becoming pregnant, and to inform the treating physician immediately should this occur.

Use in lactation: Docetaxel is a lipophilic substance but it is not known whether it is excreted in human milk. Consequently, because of the potential for adverse reactions in nursing infants, breastfeeding must be discontinued for the duration of docetaxel therapy.

## SIDE EFFECTS

These reactions were described using the NCI Common Toxicity Criteria (grade 3 = G3; grade 3 4 = G3/4; grade 4 = G4) and the COSTART terms.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

The most commonly reported adverse reactions of docetaxel alone are: neutropenia (which was reversible and not cumulative; the median day to nadir was 7 days and the median duration of severe neutropenia (< 500 cells/mm ) was 7 days), anemia, alopecia, nausea, vomiting, stomatitis, diarrhea and asthenia. The severity of adverse events of docetaxel may be increased when docetaxel is given in combination with other chemotherapeutic agents.

There was an increased incidence of SAEs and Grade 4 AEs in the trastuzumab combination arm compared to docetaxel monotherapy.

For combination with capecitabine, the most frequent treatment-related undesirable effects reported in a phase III trial in breast cancer patients failing anthracycline treatment are presented (see capecitabine summary of product characteristics).

The following adverse reactions are frequently observed with docetaxel:

## Nervous system disorders

The development of severe peripheral neurotoxicity requires a reduction of dose (see DOSAGE AND METHOD OF ADMINISTRATION and WARNINGS AND PRECAUTIONS). Mild to moderate neuro- sensory signs are characterised by paresthesia, dysesthesia or pain including burning. Neuro-motor events are mainly characterised by weakness.

## Skin and subcutaneous tissue disorders

Reversible cutaneous reactions have been observed and were generally considered as mild to moderate. Reactions were characterised by a rash including localised eruptions mainly on the feet and hands (including severe hand and foot syndrome), but also on the arms, face or thorax, and frequently associated with pruritus. Eruptions generally occurred within one week after the docetaxel infusion. Less frequently, severe symptoms such as eruptions followed by desquamation which rarely lead to interruption or discontinuation of docetaxel treatment were reported (see DOSAGE AND METHOD OF ADMINISTRATION and WARNINGS AND PRECAUTIONS). Severe nail disorders are characterised by hypo- or hyperpigmentation and sometimes pain and onycholysis.

## General disorders and administration site conditions

Infusion site reactions were generally mild and consisted of hyper pigmentation, inflammation, redness or dryness of the skin, phlebitis or extravasation and swelling of the vein.

Fluid retention includes events such as peripheral oedema and less frequently pleural effusion, pericardial efusion, ascites and weight gain. The peripheral oedema usually starts at the lower extremities and may become generalised with a weight gain of 3 kg or more. Fluid retention is cumulative in incidence and severity (see WARNINGS AND PRECAUTIONS).

## Immune system disorders

Hypersensitivity reactions have generally occurred within a few minutes following the start of the infusion of docetaxel and were usually mild to moderate. The most frequently reported symptoms were flushing, rash with or without pruritus, chest tightness, back pain, dyspnoea and fever or chills. Severe reactions were characterised by hypotension and/or bronchospasm or generalized rash/erythema (see WARNINGS AND PRECAUTIONS).

## Post-marketing experience

### Cardiac disorders

Rare cases of myocardial infarction have been reported.

### Blood and lymphatic system disorders

Bone marrow suppression and other hematologic adverse reactions have been reported. Disseminated intravascular coagulation (DIC), often in association with sepsis or multiorgan failure, has been reported.

### Nervous system disorders

Rare cases of convulsion or transient loss of consciousness have been observed with docetaxel administration. These reactions sometimes appear during the infusion of the medicinal product.

### Eye disorders

Very rare cases of transient visual disturbances (flashes, flashing lights, scotomata) typically occurring during infusion of the medicinal product and in association with hypersensitivity reactions have been reported. These were reversible upon discontinuation of the infusion.

Cases of lacrimation with or without conjunctivitis, as cases of lacrimal duct obstruction resulting in excessive tearing have been rarely reported.

### Ear and labyrinth disorders

Rare cases of ototoxicity, hearing impaired and/or hearing loss have been reported.

### Respiratory, thoracic and mediastinal disorders

Acute respiratory distress syndrome, interstitial pneumonia and pulmonary fibrosis have rarely been reported. Rare cases of radiation pneumonitis have been reported in patients receiving concomitant radiotherapy.

### Gastrointestinal disorders

Rare cases of enterocolitis, including colitis, ischemic colitis, and neutropenic enterocolitis, have been reported with a potential fatal outcome (frequency not known). Rare occurrences of dehydration as a consequence of gastrointestinal events, including enterocolitis and gastrointestinal perforation have been reported. Rare cases of ileus and intestinal obstruction have been reported.

### Skin and subcutaneous tissue disorders

Cases of cutaneous lupus erythematosus, bullous eruptions such as erythema multiforme and severe cutaneous adverse reactions such as Stevens-Johnson Syndrome (SJS), Toxic Epidermal Necrolysis(TEN) and Acute Generalized Exanthematous Pustulosis (AGEP) have been reported with docetaxel. Sclerodermal-like changes usually preceded by peripheral lymphoedema have been reported with docetaxel. Cases of permanent alopecia have (frequency not known) been reported.

### Neoplasms benign, malignant and unspecified (incl cysts and polyps)

Very rare cases of acute myeloid leukaemia and myelodysplastic syndrome have been reported in association with docetaxel when used in combination with other chemotherapy agents and/or radiotherapy.

### Vascular disorders

Venous thromboembolic events have rarely been reported.

### General disorders and administration site conditions

Radiation recall phenomena have rarely been reported. Fluid retention has not been accompanied by acute episodes of oliguria or hypotension. Dehydration and pulmonary oedema have rarely been reported.

### Immune system disorders

Some cases of anaphylactic shock, sometimes fatal, have been reported.

### Hepatobiliary disorders

Very rare cases of hepatitis, sometimes fatal primarily in patients with pre-existing liver disorders, have been reported.

## SYMPTOMS AND TREATMENT OF OVERDOSE

There is no known antidote for docetaxel overdose.

In case of overdose, the patient should be kept in a specialized unit and vital functions closely monitored.

In cases of overdose, exacerbation of adverse events may be expected. The primary anticipated complications of overdose would consist of bone marrow suppression, peripheral neurotoxicity and mucositis. Patients should receive therapeutic G-CSF as soon as possible after discovery of overdose. Other appropriate symptomatic measures should be taken, as needed.

## STORAGE CONDITION

Do not store above 25°C. Store in the original packaging in order to protect from light. Storage and shelf life after dilution: After diluted in PP infusion bags: Use immediately do not store more than 8 hours at room temperature (25°C), or more than 48 hours in refrigerator (2°C - 8°C)

Docetaxel infusion solution is supersaturated, therefore may crystallize over time. If crystals appear, the solution must no longer be used and shall be discarded.

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